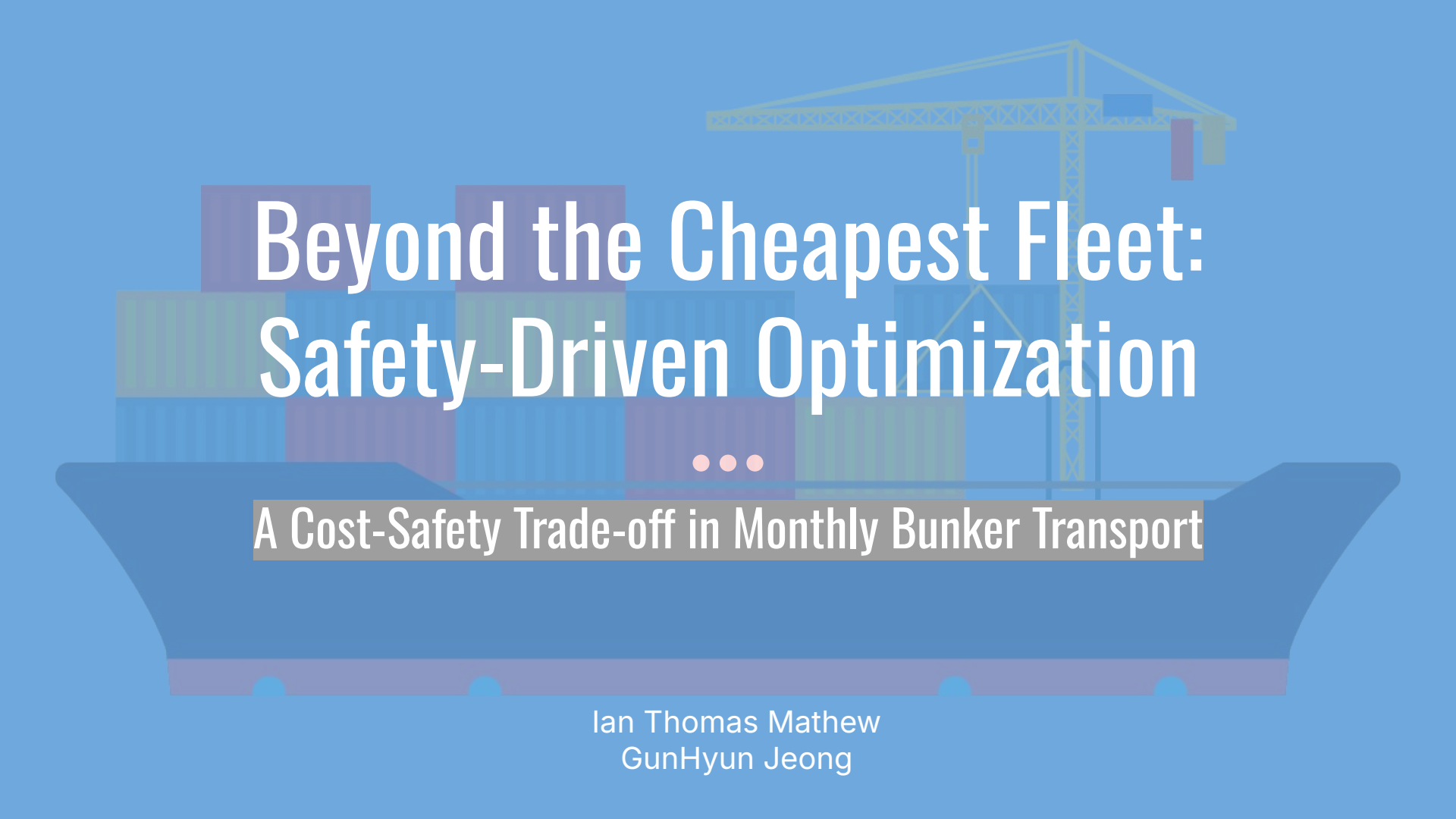


The background is a solid blue color. In the center, there is a stylized illustration of a cargo ship. The ship's hull is a darker blue. On the deck, there are several stacks of shipping containers in various shades of blue and purple. A yellow crane is positioned on the right side of the ship. Three small red dots are visible on the ship's deck, between the container stacks.

Beyond the Cheapest Fleet: Safety-Driven Optimization

A Cost-Safety Trade-off in Monthly Bunker Transport

Ian Thomas Mathew
GunHyun Jeong

The Fleet Selection Problem

1. Moves monthly bunker cargo
2. Minimizes total system cost
3. Meets safety and fuel diversity
4. Uses each vessel only once



Core Challenge: Balancing cost, safety, and emission in one decision



Two Safety Scenarios

Baseline Fleet

- Average Safety ≥ 3
- Cost optimal under standard conditions

High-safety fleet

- Average Safety ≥ 4
- Represents stricter industry expectations

Baseline	High-Safety
Safety ≥ 3	Safety ≥ 4
Cost focus	Safety focus

Goal: Understand how higher safety changes cost and fleet structure



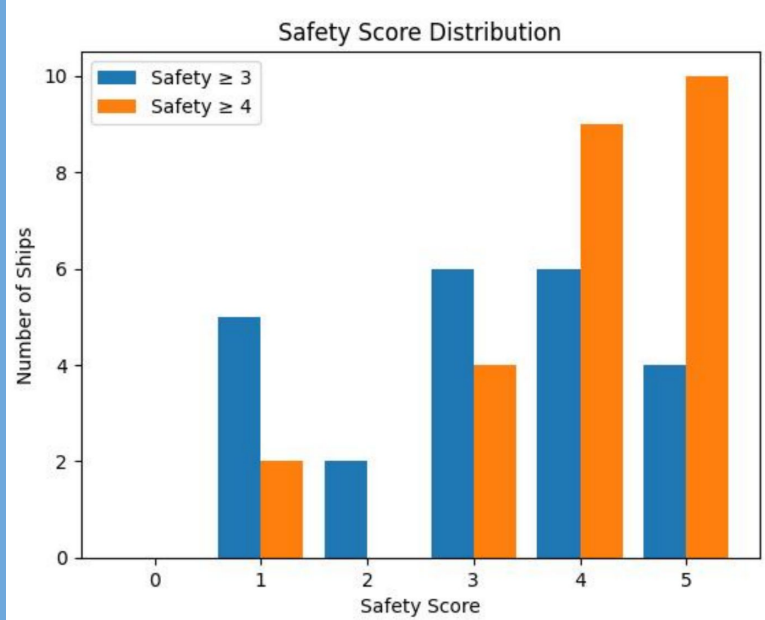
Baseline vs High-Safety Fleet

Metric	Safety ≥ 3	Safety ≥ 4
Fleet size	23 ships	25 ships
Total cost	\$1.53B	\$1.69B
Avg safety	3.09	4.00
Emissions	56k t CO ₂ e	52k t CO ₂ e

Key takeaway: +10% cost → significantly safer fleet with lower emissions



What Changes When Safety Increases?



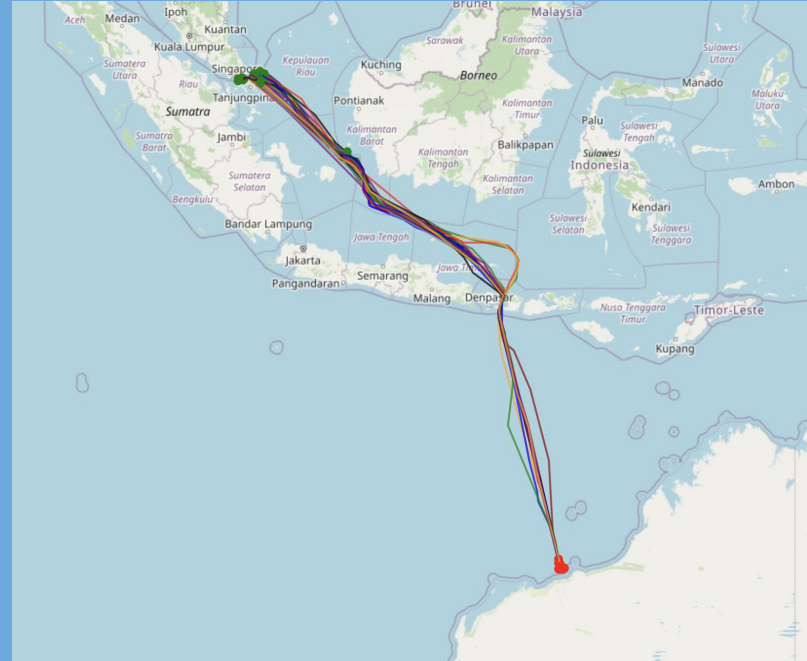
- Low-safety vessels
- + High-safety vessels
- + Fleet size

Insight: Safety acts as a binding constraint in the model



Operational Realism and Final Takeaways

Selected vessels follow real AIS-derived routes between Singapore and Australia



Final takeaway:

Higher safety standards raise costs, but also reduce emissions and operational risk

